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| | **ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)** | | **Assignment 1** |
| Branch | CSIT | Programme | **B.Tech** |
| Course Name | PROGRAMMING PROJECTS WITH JAVASCRIPT(PPWJ) | Semester | 5th |
| Course Code | CSE 3838 | Academic Year | 2025-2026 |
| **Lab Assignment: 1** | | **Topic: Setting up a project with Vite & ReactJs** | |

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| Learning Level (LL) | **L1**: Remembering | **L3:** Applying | **L5**: Evaluating |
| | **L2**: Understanding | **L4**: Analysing | **L6**: Creating |

| Q's | Questions | COs | LL |
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| **1.** | Create a simple Hello World React application using Vite. The application should consist of a functional component that returns JSX displaying a heading element. Create a Vite React project and explain what happens when you run npm create vite@latest. Explain the role of package.json. | **CO1** | **L2 L3** |
| **2.** | Create a simple counter application using React and Vite. Perform the following tasks.<br>• Create a React component named Counter.<br>• Initialize a state variable to store the counter value.<br>• Display the current counter value on the screen.<br>• Add two buttons:<br>(i) One to increment the counter<br>(ii) One to decrement the counter<br>• Update the UI dynamically when the button is clicked. | **CO1** | **L3 L4** |
| **3.** | Develop a theme toggle feature that switches between Light Mode and Dark Mode. Perform the following tasks.<br>• Create a React component called ThemeToggle.<br>• Use a boolean state variable to track the current theme.<br>• Change background color and text color based on the selected theme.<br>• Display the current mode (Light/Dark) on the screen.<br>• Add a button to toggle between themes.<br>**Expected Outcome**<br>Clicking the toggle button should switch the UI between light and dark themes. | **CO1** | **L3,L4** |
| **4.** | Demonstrate the use of props to create reusable components. | **CO1** | **L3, L4** |

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| | • Create a Greeting component that accepts a name prop.<br>• Display a greeting message using the passed prop.<br>• Reuse the component multiple times with different names.<br>**Expected Outcome**<br>Different greeting messages should be displayed for each name passed as a prop. | | |
| **5** | Implement navigation between screen using React Router with Vite.<br>Tasks<br>• Create at least two pages (Home, About).<br>• Configure routes in the main application.<br>• Add navigation links between pages.<br>• Ensure page content changes without reloading.<br>**Expected Outcome**<br>Users should be able to navigate between pages smoothly. | **CO1** | **L3,L4** |

1. Assignment carries a weightage of 20 marks out of 100
2. Only CO1 outcomes are covered.

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| Course Outcomes | CO1 | Understand and configure a modern full-stack development environment using Node.js, MongoDB, Docker, and supporting tools. |
| | CO2 | Design and implement RESTful backend services using Express, MongoDB (Mongoose), and unit testing frameworks. |
| | CO3 | Develop interactive frontend applications using React and efficiently integrate them with backend APIs using modern data-fetching techniques. |
| | CO4 | Apply DevOps practices to containerize, test, and deploy full-stack applications using Docker and CI/CD pipelines. |
| | CO5 | Implement authentication, performance optimization, Search Engine Optimization techniques, and automated testing for scalable and secure web applications. |
| | CO6 | Build and integrate advanced full-stack architectures using data aggregation, visualization, and GraphQL APIs with modern frontend clients. |